CO4 AT - Question 58

Maximum Number of Non-Overlapping Segments using Greedy Strategy

Aim:
To select the maximum number of non-overlapping segments using the Greedy strategy.

Problem Statement:
Given a set of segments, select the maximum number of non-overlapping segments such that the count of selected segments is maximized. Use a Greedy approach to solve the problem efficiently.

Input:
n = 3
Segments = (1,2), (2,3), (3,4)

Algorithm:
1. Sort all segments based on their ending times.
2. Select the segment that ends earliest.
3. Continue selecting the next segment whose start time is greater than or equal to the end time of the previously selected segment.
4. Count the selected segments.

Greedy Strategy:
Always choose the segment with the earliest finishing time to maximize the total number of non-overlapping segments.

Expected Output:
Max Segments = 3

Time Complexity:
O(n log n)

Space Complexity:
O(n)

Result:
The maximum number of non-overlapping segments was successfully determined using the Greedy approach. For the given input, the maximum number of segments selected was 3.

Conclusion:
The Greedy algorithm efficiently solves the problem by always selecting the earliest finishing segment, thereby maximizing the number of non-overlapping segments.

Output Screenshot:
(Insert terminal output screenshot here.)